COMP122 · DATABASE SYSTEMS PROJECT

Enterprise Car Rental Database

A relational database design — from stakeholder requirements through normalized schema to live Oracle implementation.

01 — SYSTEM OVERVIEW

What the Database Powers

A single relational schema supporting the full rental lifecycle for a multi-branch car rental enterprise — from search to settlement.

Vehicle Inventory

Fleet, categories & manufacturers across branches

Reservations

Booking lifecycle from search to pickup

Customer Management

Profiles, licenses & loyalty tiers

Logistics

Inter-branch vehicle transfers

Supply Chain

Vendor procurement & parts tracking

Financial Reporting

Revenue, utilization & segmentation

Four Roles, Four Views

We identified four key stakeholders and mapped a separate external view for each, so every user sees only the data relevant to their role.



Customers

- Search & book vehicles
- Manage their profile
- Save vehicle wishlists



Rental Agents

- Process transactions
- Handle check-ins & returns
- Verify driver's licenses



Fleet Managers

- Track inter-branch transfers
- Schedule maintenance



Executives

- Monthly revenue reports
- Utilization metrics
- Customer segmentation

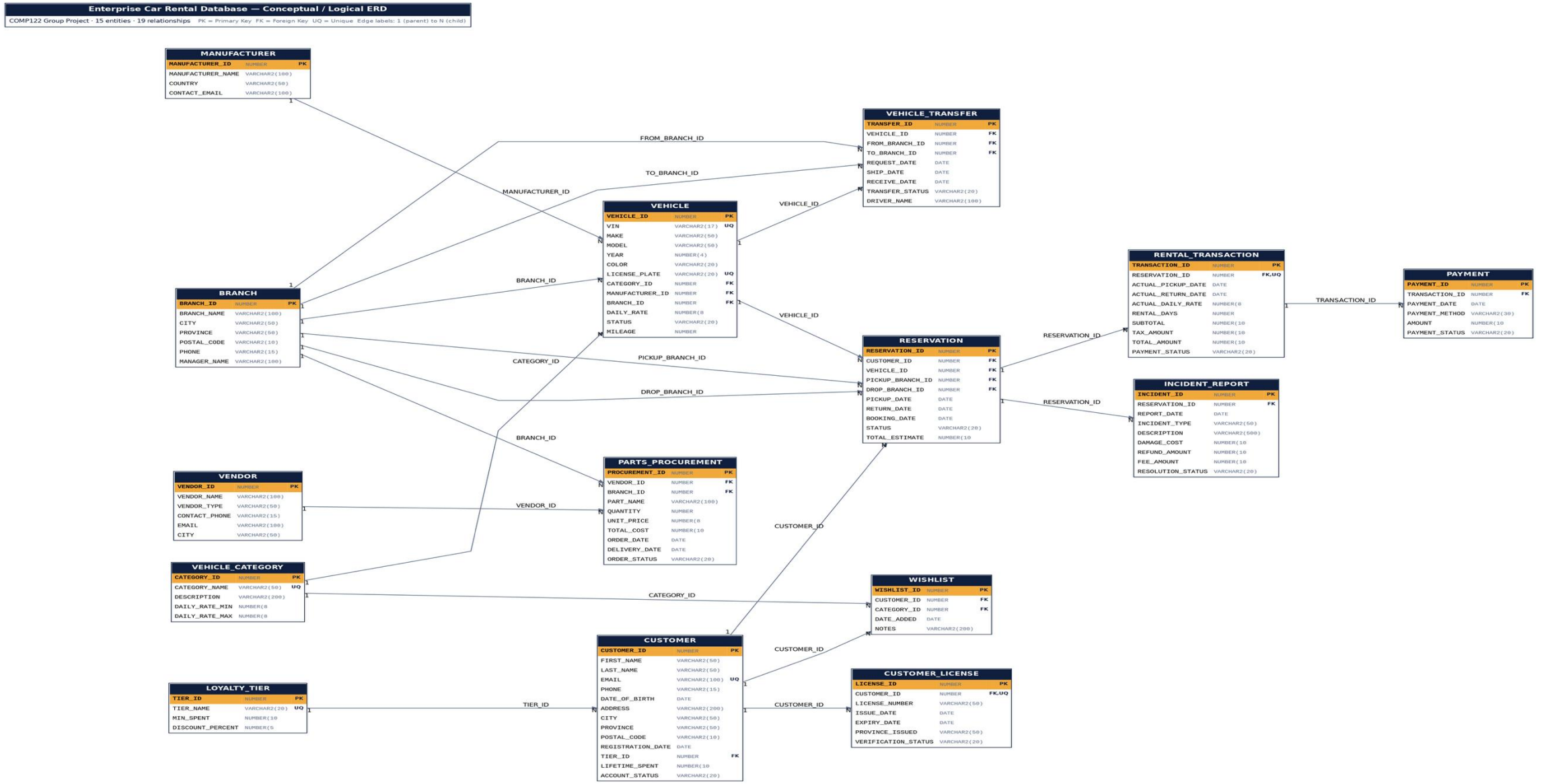
15 Entities, One Consistent Model

15	4	3NF
Entities	Core Entities	Normal Form

BRANCH	VEHICLE	CUSTOMER	RESERVATION
VEHICLE_TRANSFER	VEHICLE_CATEGORY	CUSTOMER_LICENSE	RENTAL_TRANSACTION
VENDOR	MANUFACTURER	WISHLIST	PAYMENT
PARTS_PROCUREMENT		LOYALTY_TIER	INCIDENT_REPORT

Key relationships: BRANCH has many VEHICLE · CUSTOMER makes many RESERVATION · RESERVATION generates one RENTAL_TRANSACTION · VEHICLE can transfer between BRANCH records. All relationships are fully normalized to Third Normal Form (3NF).

Every Table, Every Relationship



Enforcing Integrity at the Database Level

Primary Keys

Uniquely identify every row

Foreign Keys

Enforce valid relationships

Unique & Not Null

Block duplicate or missing data

Check Constraints

Validate business rules on write

Example CHECK constraint

```
RESERVATION: RETURN_DATE must be strictly after PICKUP_DATE
```

VARCHAR2	NUMBER	DATE
Text	Currency & IDs	Temporal data

SCHEMA

```
Name                 Null?    Type
-------------------- -------- ------------
RESERVATION_ID       NOT NULL NUMBER
CUSTOMER_ID          NOT NULL NUMBER
VEHICLE_ID           NOT NULL NUMBER
PICKUP_BRANCH_ID     NOT NULL NUMBER
DROP_BRANCH_ID       NOT NULL NUMBER
PICKUP_DATE          NOT NULL DATE
RETURN_DATE          NOT NULL DATE
BOOKING_DATE                  DATE
STATUS                        VARCHAR2(20)
TOTAL_ESTIMATE                NUMBER(10,2)
```

RESERVATION table structure — DESC output

15 Tables, Live in Oracle

15

Tables Created

10+

Sample Rows per Table

Every table was created and populated directly in our Oracle schema, ready to support all ten business queries.

VERIFY

```
SELECT table_name FROM user_tables WHERE table_name IN (
'BRANCH','VEHICLE_CATEGORY','MANUFACTURER','VEHICLE','CUSTOMER',
'CUSTOMER_LICENSE','RESERVATION','RENTAL_TRANSACTION','PAYMENT',
'VEHICLE_TRANSFER','VENDOR','PARTS_PROCUREMENT','WISHLIST',
'INCIDENT_REPORT','LOYALTY_TIER'
) ORDER BY table_name;
```

SCHEMA

	TABLE_NAME
1	BRANCH
2	CUSTOMER
3	CUSTOMER_LICENSE
4	INCIDENT_REPORT
5	LOYALTY_TIER
6	MANUFACTURER
7	PARTS_PROCUREMENT
8	PAYMENT
9	RENTAL_TRANSACTION
10	RESERVATION
11	VEHICLE
12	VEHICLE_CATEGORY
13	VEHICLE_TRANSFER
14	VENDOR
15	WISHLIST

All 15 Tables

- BRANCH
- VEHICLE_CATEGORY
- MANUFACTURER
- VEHICLE
- CUSTOMER
- CUSTOMER_LICENSE
- RESERVATION
- RENTAL_TRANSACTION
- PAYMENT
- VEHICLE_TRANSFER
- VENDOR
- PARTS_PROCUREMENT
- WISHLIST
- INCIDENT_REPORT
- LOYALTY_TIER

10

Business Queries

Each query answers a real operational question — from vehicle search to executive reporting.

01 Vehicle Discovery

02 Reservation Management

03 User Account Management

04 Logistics Tracking

05 Supply Chain Management

06 Executive Reporting

07 Inventory Analytics

08 Customer Segmentation

09 Wish List Engagement

10 Customer Service Processing

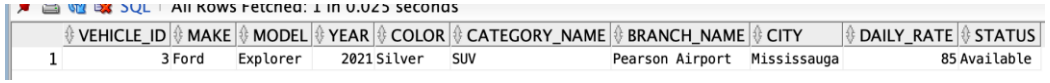
Vehicle Discovery

Filters available SUVs under $100/day in Mississauga, joining VEHICLE, VEHICLE_CATEGORY, and BRANCH.

SQL QUERY

```sql
SELECT
    v.VEHICLE_ID,
    v.MAKE,
    v.MODEL,
    v.YEAR,
    v.COLOR,
    vc.CATEGORY_NAME,
    b.BRANCH_NAME,
    b.CITY,
    v.DAILY_RATE,
    v.STATUS
FROM VEHICLE v
JOIN VEHICLE_CATEGORY vc ON v.CATEGORY_ID = vc.CATEGORY_ID
JOIN BRANCH b ON v.BRANCH_ID = b.BRANCH_ID
WHERE v.STATUS = 'Available'
    AND vc.CATEGORY_NAME = 'SUV'
    AND b.CITY = 'Mississauga'
    AND v.DAILY_RATE <= 100.00
ORDER BY v.DAILY_RATE;
```

QUERY RESULT



All Rows Fetched: 1 in 0.025 seconds

	VEHICLE_ID	MAKE	MODEL	YEAR	COLOR	CATEGORY_NAME	BRANCH_NAME	CITY	DAILY_RATE	STATUS
1	3	Ford	Explorer	2021	Silver	SUV	Pearson Airport	Mississauga	85	Available

Reservation Management

Tracks every active and confirmed booking with customer, vehicle, and branch details across four joins.

SQL QUERY

```sql
SELECT
    r.RESERVATION_ID,
    c.FIRST_NAME || ' ' || c.LAST_NAME AS CUSTOMER_NAME,
    v.MAKE || ' ' || v.MODEL AS VEHICLE,
    b1.BRANCH_NAME AS PICKUP_BRANCH,
    b2.BRANCH_NAME AS DROP_BRANCH,
    r.PICKUP_DATE,
    r.RETURN_DATE,
    r.STATUS,
    r.TOTAL_ESTIMATE
FROM RESERVATION r
JOIN CUSTOMER c ON r.CUSTOMER_ID = c.CUSTOMER_ID
JOIN VEHICLE v ON r.VEHICLE_ID = v.VEHICLE_ID
JOIN BRANCH b1 ON r.PICKUP_BRANCH_ID = b1.BRANCH_ID
JOIN BRANCH b2 ON r.DROP_BRANCH_ID = b2.BRANCH_ID
WHERE r.STATUS IN ('Confirmed','Active')
ORDER BY r.PICKUP_DATE;
```

QUERY RESULT

	RESERVATION_ID	CUSTOMER_NAME	VEHICLE	PICKUP_BRANCH	DROP_BRANCH	PICKUP_DATE	RETURN_DATE	STATUS	TOTAL_ESTIMATE
1	1	John Doe	Honda Civic	Downtown Toronto	Pearson Airport	26-08-31	26-09-04	Active	260
2	9	David Garcia	Toyota RAV4	Winnipeg Central	Winnipeg Central	26-09-11	26-09-17	Active	570
3	7	Linda Taylor	Ford Mustang	Pearson Airport	Pearson Airport	26-09-19	26-09-21	Active	320
4	12	Patricia Jackson	Hyundai Tucson	Montreal Old Port	Edmonton South	26-09-24	26-09-29	Active	375
5	3	Jane Smith	Ford Explorer	Vancouver Central	Vancouver Central	26-10-01	26-10-07	Confirmed	510
6	4	Chris Wilson	Volkswagen Golf	Ottawa Rideau	Halifax Waterfront	26-10-05	26-10-10	Confirmed	250
7	10	James Anderson	Honda Accord	Quebec City Centre	Quebec City Centre	26-10-15	26-10-20	Confirmed	350
8	6	Laura Martinez	Nissan Leaf	Ottawa Rideau	Ottawa Rideau	26-10-31	26-11-04	Confirmed	280
9	11	Robert Thomas	Tesla Model 3	Downtown Toronto	Downtown Toronto	26-11-09	26-11-14	Confirmed	400
10	8	Emily Davis	Toyota Corolla	Montreal Old Port	Montreal Old Port	26-12-01	26-12-10	Confirmed	495

User Account Management

Verifies driver's licenses and calculates rental eligibility in real time with a LEFT JOIN and CASE.

SQL QUERY

```sql
SELECT
    c.CUSTOMER_ID,
    c.FIRST_NAME,
    c.LAST_NAME,
    c.EMAIL,
    c.PHONE,
    c.ADDRESS,
    c.ACCOUNT_STATUS,
    cl.LICENSE_NUMBER,
    cl.EXPIRY_DATE,
    cl.VERIFICATION_STATUS,
    CASE
        WHEN cl.VERIFICATION_STATUS = 'Verified' THEN 'Approved for Rental'
        WHEN cl.VERIFICATION_STATUS = 'Pending' THEN 'Awaiting Verification'
        WHEN cl.VERIFICATION_STATUS = 'Expired' THEN 'License Expired - Action Required'
        ELSE 'Not Eligible'
    END AS RENTAL_ELIGIBILITY
FROM CUSTOMER c
LEFT JOIN CUSTOMER_LICENSE cl ON c.CUSTOMER_ID = cl.CUSTOMER_ID
ORDER BY c.CUSTOMER_ID;
```

QUERY RESULT

SQL | All Rows Fetched: 12 in 0.175 seconds

	CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	ADDRESS	ACCOUNT_STATUS	LICENSE_NUMBER	EXPIRY_DATE	VERIFICATION_STATUS	RENTAL_ELIGIBILITY
1	1	John	Doe	john.doe@email.com	416-555-1001	123 King St	Active	D1234567890	29-11-30	Verified	Approved for Rental
2	2	Jane	Smith	jane.smith@email.com	604-555-1002	456 Robson St	Active	D0987654321	30-09-14	Verified	Approved for Rental
3	3	Michael	Brown	michael.b@email.com	403-555-1003	789 4 Ave SW	Active	D1122334455	29-05-21	Verified	Approved for Rental
4	4	Emily	Davis	emily.d@email.com	514-555-1004	321 Old Port Rd	Active	D5566778899	27-07-12	Pending	Awaiting Verification
5	5	Chris	Wilson	chris.w@email.com	613-555-1005	654 Rideau St	Active	D9988776655	30-03-06	Verified	Approved for Rental
6	6	Sarah	Miller	sarah.m@email.com	780-555-1006	987 Whyte Ave	Active	D3344556677	30-11-17	Verified	Approved for Rental
7	7	David	Garcia	david.g@email.com	902-555-1007	147 Waterfront Dr	Active	D7766554433	32-01-21	Verified	Approved for Rental
8	8	Laura	Martinez	laura.m@email.com	204-555-1008	258 Portage Ave	Active	D4455667788	30-06-01	Expired	License Expired - Action Required
9	9	James	Anderson	james.a@email.com	418-555-1009	369 Champlain Blvd	Active	D2233445566	31-10-14	Verified	Approved for Rental
10	10	Linda	Taylor	linda.t@email.com	416-555-1010	753 Bay St	Active	D6677889900	31-03-01	Verified	Approved for Rental
11	11	Robert	Thomas	robert.t@email.com	905-555-1011	159 Dundas St	Active	D8899001122	30-07-21	Pending	Awaiting Verification
12	12	Patricia	Jackson	patricia.j@email.com	604-555-1012	357 Granville St	Active	D3344221155	29-12-09	Verified	Approved for Rental

Logistics Tracking

Shows every vehicle transfer between branches, its fulfillment status, and the assigned driver.

SQL QUERY

```sql
SELECT
    vt.TRANSFER_ID,
    v.MAKE || ' ' || v.MODEL AS VEHICLE,
    v.VIN,
    b1.BRANCH_NAME AS FROM_BRANCH,
    b1.CITY AS FROM_CITY,
    b2.BRANCH_NAME AS TO_BRANCH,
    b2.CITY AS TO_CITY,
    vt.REQUEST_DATE,
    vt.SHIP_DATE,
    vt.RECEIVE_DATE,
    vt.TRANSFER_STATUS,
    vt.DRIVER_NAME,
    CASE
        WHEN vt.TRANSFER_STATUS = 'Fulfilled' THEN 'Completed'
        WHEN vt.TRANSFER_STATUS = 'In Transit' THEN 'On The Way'
        WHEN vt.TRANSFER_STATUS = 'Requested' THEN 'Awaiting Dispatch'
        ELSE 'Cancelled'
    END AS FULFILLMENT_STATUS
FROM VEHICLE_TRANSFER vt
JOIN VEHICLE v ON vt.VEHICLE_ID = v.VEHICLE_ID
JOIN BRANCH b1 ON vt.FROM_BRANCH_ID = b1.BRANCH_ID
JOIN BRANCH b2 ON vt.TO_BRANCH_ID = b2.BRANCH_ID
ORDER BY vt.REQUEST_DATE DESC;
```

QUERY RESULT

	TRANSFER_ID	VEHICLE	VIN	FROM_BRANCH	FROM_CITY	TO_BRANCH	TO_CITY	REQUEST_DATE	SHIP_DATE	RECEIVE_DATE	TRANSFER_STATUS	DRIVER_NAME	FULFILLMENT_STATUS
1	7	Ford Mustang	1HGCM82633A123468	Pearson Airport	Mississauga	Calgary Downtown	Calgary	26-10-01	(null)	(null)	Requested	(null)	Awaiting Dispatch
2	8	Toyota Corolla	1HGCM82633A123456	Downtown Toronto	Toronto	Ottawa Rideau	Ottawa	26-09-19	26-09-20	(null)	In Transit	Bruce Wayne	On The Way
3	10	Tesla Model Y	1HGCM82633A123470	Calgary Downtown	Calgary	Montreal Old Port	Montreal	26-09-14	26-09-15	(null)	In Transit	Peter Parker	On The Way
4	5	Ford Explorer	1HGCM82633A123458	Pearson Airport	Mississauga	Montreal Old Port	Montreal	26-09-09	26-09-10	(null)	In Transit	Ellen Ripley	On The Way
5	3	BMW i4	1HGCM82633A123469	Vancouver Central	Vancouver	Calgary Downtown	Calgary	26-09-04	(null)	(null)	Requested	(null)	Awaiting Dispatch
6	2	BMW X5	1HGCM82633A123459	Vancouver Central	Vancouver	Downtown Toronto	Toronto	26-08-31	26-09-01	(null)	In Transit	Sarah Connor	On The Way
7	9	Honda Accord	1HGCM82633A123467	Quebec City Centre	Quebec City	Halifax Waterfront	Halifax	26-08-26	26-08-28	26-08-31	Fulfilled	Clark Kent	Completed
8	4	Chevrolet Silverado	1HGCM82633A123464	Edmonton South	Edmonton	Pearson Airport	Mississauga	26-08-11	26-08-13	26-08-15	Fulfilled	John Wick	Completed
9	1	Hyundai Tucson	1HGCM82633A123462	Montreal Old Port	Montreal	Vancouver Central	Vancouver	26-08-02	26-08-04	26-08-06	Fulfilled	Mike Ross	Completed
10	6	Toyota RAV4	1HGCM82633A123466	Winnipeg Central	Winnipeg	Downtown Toronto	Toronto	26-07-16	26-07-18	26-07-20	Fulfilled	Tony Stark	Completed

Supply Chain Management

Displays vendor procurement records and calculates delivery days to measure vendor performance.

SQL QUERY

```sql
SELECT
    p.PROCUREMENT_ID,
    v.VENDOR_NAME,
    v.VENDOR_TYPE,
    b.BRANCH_NAME,
    p.PART_NAME,
    p.QUANTITY,
    p.UNIT_PRICE,
    p.TOTAL_COST,
    p.ORDER_DATE,
    p.DELIVERY_DATE,
    p.ORDER_STATUS,
    CASE
        WHEN p.DELIVERY_DATE IS NOT NULL THEN p.DELIVERY_DATE - p.ORDER_DATE
        ELSE NULL
    END AS DELIVERY_DAYS
FROM PARTS_PROCUREMENT p
JOIN VENDOR v ON p.VENDOR_ID = v.VENDOR_ID
JOIN BRANCH b ON p.BRANCH_ID = b.BRANCH_ID
ORDER BY p.ORDER_DATE DESC;
```

QUERY RESULT

	PROCUREMENT_ID	VENDOR_NAME	VENDOR_TYPE	BRANCH_NAME	PART_NAME	QUANTITY	UNIT_PRICE	TOTAL_COST	ORDER_DATE	DELIVERY_DATE	ORDER_STATUS	DELIVERY_DAYS
1	6	MechMaster	Service	Montreal Old Port	Engine Diagnostics	30	75	2250	26-09-09	(null)	Ordered	(null)
2	10	RoadGuard Insurance	Insurance	Quebec City Centre	Claim Processing	5	200	1000	26-09-04	(null)	Shipped	(null)
3	3	BatteryWorld	Parts	Ottawa Rideau	Car Batteries	10	150	1500	26-08-31	(null)	Shipped	(null)
4	5	GlassPro Auto	Parts	Calgary Downtown	Windshield Glass	5	300	1500	26-08-21	(null)	Ordered	(null)
5	9	LubeExpress	Service	Winnipeg Central	Lube Service Package	25	40	1000	26-08-11	26-08-21	Delivered	10
6	2	TireMax Canada	Parts	Vancouver Central	All Season Tires	16	120	1920	26-08-06	26-08-16	Delivered	10
7	7	EcoParts Supply	Parts	Edmonton South	Eco Filters	40	25	1000	26-07-26	26-08-06	Delivered	11
8	4	FleetFix Services	Service	Pearson Airport	Oil Change Service	50	35	1750	26-07-16	26-07-26	Delivered	10
9	1	AutoParts Plus	Parts	Downtown Toronto	Brake Pads	20	45	900	26-07-11	26-07-21	Delivered	10
10	8	Shield Insurance	Insurance	Halifax Waterfront	Insurance Policy Bundle	1	5000	5000	26-07-02	26-07-02	Delivered	0

Executive Reporting

Calculates monthly rental revenue, average revenue per rental, and fleet utilization rate.

SQL QUERY

```sql
SELECT
    TO_CHAR(rt.ACTUAL_PICKUP_DATE,'YYYY-MM') AS MONTH,
    COUNT(DISTINCT rt.RESERVATION_ID) AS TOTAL_RENTALS,
    SUM(rt.TOTAL_AMOUNT) AS TOTAL_REVENUE,
    AVG(rt.TOTAL_AMOUNT) AS AVG_REVENUE_PER_RENTAL,
    SUM(rt.RENTAL_DAYS) AS TOTAL_RENTAL_DAYS,
    COUNT(DISTINCT r.VEHICLE_ID) AS UNIQUE_VEHICLES_USED,
    ROUND(SUM(rt.RENTAL_DAYS) / (COUNT(DISTINCT r.VEHICLE_ID) * 30) * 100, 2) AS UTILIZATION_RATE_PERCENT
FROM RENTAL_TRANSACTION rt
JOIN RESERVATION r ON rt.RESERVATION_ID = r.RESERVATION_ID
WHERE rt.PAYMENT_STATUS = 'Paid'
GROUP BY TO_CHAR(rt.ACTUAL_PICKUP_DATE,'YYYY-MM')
ORDER BY MONTH;
```

QUERY RESULT

	MONTH	TOTAL_RENTALS	TOTAL_REVENUE	AVG_REVENUE_PER_RENTAL	TOTAL_RENTAL_DAYS	UNIQUE_VEHICLES_USED	UTILIZATION_RATE_PERCENT
1	2026-08	1	565	565	5	1	16.67
2	2026-09	1	847.5	847.5	5	1	16.67

Inventory Analytics

Ranks the most popular vehicle models by rental volume using the RANK analytic function.

SQL QUERY

```sql
SELECT
    v.MAKE,
    v.MODEL,
    vc.CATEGORY_NAME,
    COUNT(r.RESERVATION_ID) AS TOTAL_RENTALS,
    SUM(rt.TOTAL_AMOUNT) AS TOTAL_REVENUE,
    RANK() OVER (ORDER BY COUNT(r.RESERVATION_ID) DESC) AS POPULARITY_RANK
FROM VEHICLE v
JOIN VEHICLE_CATEGORY vc ON v.CATEGORY_ID = vc.CATEGORY_ID
LEFT JOIN RESERVATION r ON v.VEHICLE_ID = r.VEHICLE_ID
LEFT JOIN RENTAL_TRANSACTION rt ON r.RESERVATION_ID = rt.RESERVATION_ID
GROUP BY v.MAKE, v.MODEL, vc.CATEGORY_NAME
ORDER BY TOTAL_RENTALS DESC;
```

QUERY RESULT

	MAKE	MODEL	CATEGORY_NAME	TOTAL_RENTALS	TOTAL_REVENUE	POPULARITY_RANK
1	Nissan	Leaf	Electric	1	(null)	1
2	Tesla	Model 3	Electric	1	(null)	1
3	Honda	Accord	Sedan	1	(null)	1
4	Toyota	RAV4	SUV	1	(null)	1
5	Tesla	Model Y	SUV	1	565	1
6	Ford	Mustang	Sports	1	(null)	1
7	Ford	Explorer	SUV	1	(null)	1
8	Mercedes	C-Class	Luxury	1	847.5	1
9	Volkswagen	Golf	Economy	1	(null)	1
10	Hyundai	Tucson	SUV	1	(null)	1
11	Honda	Civic	Sedan	1	(null)	1
12	Toyota	Corolla	Economy	1	(null)	1
13	Chevrolet	Silverado	Truck	0	(null)	13
14	BMW	X5	SUV	0	(null)	13
15	BMW	i4	Electric	0	(null)	13

Customer Segmentation

Classifies customers into Bronze / Silver / Gold / VIP tiers and validates stored vs. calculated tier.

SQL QUERY

```sql
SELECT
    c.CUSTOMER_ID,
    c.FIRST_NAME,
    c.LAST_NAME,
    c.EMAIL,
    c.LIFETIME_SPENT,
    lt.TIER_NAME,
    lt.DISCOUNT_PERCENT || '%' AS TIER_DISCOUNT,
    CASE
        WHEN c.LIFETIME_SPENT >= 5000 THEN 'VIP'
        WHEN c.LIFETIME_SPENT >= 2000 THEN 'Gold'
        WHEN c.LIFETIME_SPENT >= 500 THEN 'Silver'
        ELSE 'Bronze'
    END AS CALCULATED_TIER,
    COUNT(r.RESERVATION_ID) AS TOTAL_RENTALS
FROM CUSTOMER c
JOIN LOYALTY_TIER lt ON c.TIER_ID = lt.TIER_ID
LEFT JOIN RESERVATION r ON c.CUSTOMER_ID = r.CUSTOMER_ID
GROUP BY c.CUSTOMER_ID, c.FIRST_NAME, c.LAST_NAME, c.EMAIL,
         c.LIFETIME_SPENT, lt.TIER_NAME, lt.DISCOUNT_PERCENT
ORDER BY c.LIFETIME_SPENT DESC;
```

QUERY RESULT

	CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	LIFETIME_SPENT	TIER_NAME	TIER_DISCOUNT	CALCULATED_TIER	TOTAL_RENTALS
1	10	Linda	Taylor	linda.t@email.com	7500	VIP	15%	VIP	1
2	6	Sarah	Miller	sarah.m@email.com	6000	VIP	15%	VIP	1
3	3	Michael	Brown	michael.b@email.com	2500	Gold	10%	Gold	1
4	8	Laura	Martinez	laura.m@email.com	2200	Gold	10%	Gold	1
5	2	Jane	Smith	jane.smith@email.com	1200	Silver	5%	Silver	1
6	9	James	Anderson	james.a@email.com	1100	Silver	5%	Silver	1
7	5	Chris	Wilson	chris.w@email.com	800	Silver	5%	Silver	1
8	1	John	Doe	john.doe@email.com	350	Bronze	0%	Bronze	1
9	11	Robert	Thomas	robert.t@email.com	200	Bronze	0%	Bronze	1
10	4	Emily	Davis	emily.d@email.com	150	Bronze	0%	Bronze	1
11	7	David	Garcia	david.g@email.com	75	Bronze	0%	Bronze	1
12	12	Patricia	Jackson	patricia.j@email.com	50	Bronze	0%	Bronze	1

Wish List Engagement

Extracts active wishlist items and ranks each customer's most recent saves with DENSE_RANK.

SQL QUERY

```sql
SELECT
    c.CUSTOMER_ID,
    c.FIRST_NAME || ' ' || c.LAST_NAME AS CUSTOMER_NAME,
    c.EMAIL,
    vc.CATEGORY_NAME AS WISHLIST_VEHICLE_CLASS,
    vc.DESCRIPTION,
    w.DATE_ADDED,
    w.NOTES,
    DENSE_RANK() OVER (PARTITION BY c.CUSTOMER_ID ORDER BY w.DATE_ADDED DESC) AS WISH_PRIORITY
FROM WISHLIST w
JOIN CUSTOMER c ON w.CUSTOMER_ID = c.CUSTOMER_ID
JOIN VEHICLE_CATEGORY vc ON w.CATEGORY_ID = vc.CATEGORY_ID
WHERE c.ACCOUNT_STATUS = 'Active'
ORDER BY c.CUSTOMER_ID, w.DATE_ADDED DESC;
```

QUERY RESULT

	CUSTOMER_ID	CUSTOMER_NAME	EMAIL	WISHLIST_VEHICLE_CLASS	DESCRIPTION	DATE_ADDED	NOTES	WISH_PRIORITY
1	1	John Doe	john.doe@email.com	SUV	Sport utility vehicles	26-07-16	Need SUV for family trip	1
2	2	Jane Smith	jane.smith@email.com	Electric	Battery electric vehicles	26-08-11	Want to try electric	1
3	3	Michael Brown	michael.b@email.com	Luxury	Premium and luxury vehicles	26-07-21	Luxury for anniversary	1
4	4	Emily Davis	emily.d@email.com	Sedan	Mid-size family sedans	26-08-31	Sedan for commute	1
5	5	Chris Wilson	chris.w@email.com	Truck	Pickup trucks	26-08-16	Truck for moving	1
6	6	Sarah Miller	sarah.m@email.com	Sports	High performance sports cars	26-07-06	Sports car dream	1
7	7	David Garcia	david.g@email.com	Van	Passenger and cargo vans	26-09-09	Van for group trip	1
8	8	Laura Martinez	laura.m@email.com	Hybrid	Hybrid electric vehicles	26-08-21	Eco-friendly option	1
9	9	James Anderson	james.a@email.com	Convertible	Open-top sports cars	26-09-04	Summer convertible	1
10	10	Linda Taylor	linda.t@email.com	SUV	Sport utility vehicles	26-07-26	Spacious SUV	1
11	11	Robert Thomas	robert.t@email.com	Economy	Compact cars, fuel efficient	26-10-01	Budget friendly	1
12	12	Patricia Jackson	patricia.j@email.com	Electric	Battery electric vehicles	26-10-05	Electric for city	1

Customer Service Processing

Calculates final billing from rental cost, damage, and fees, using NVL to safely handle nulls.

SQL QUERY

```sql
SELECT
    r.RESERVATION_ID,
    c.FIRST_NAME || ' ' || c.LAST_NAME AS CUSTOMER_NAME,
    v.MAKE || ' ' || v.MODEL AS VEHICLE,
    rt.ACTUAL_RETURN_DATE AS RETURN_DATE,
    ir.INCIDENT_TYPE,
    ir.DESCRIPTION,
    ir.DAMAGE_COST,
    ir.FEE_AMOUNT,
    ir.REFUND_AMOUNT,
    (NVL(ir.DAMAGE_COST,0) + NVL(ir.FEE_AMOUNT,0) - NVL(ir.REFUND_AMOUNT,0)) AS NET_CHARGE,
    rt.TOTAL_AMOUNT AS ORIGINAL_RENTAL_COST,
    (rt.TOTAL_AMOUNT + NVL(ir.DAMAGE_COST,0) + NVL(ir.FEE_AMOUNT,0) - NVL(ir.REFUND_AMOUNT,0)) AS FINAL_BILL,
    ir.RESOLUTION_STATUS
FROM RESERVATION r
JOIN CUSTOMER c ON r.CUSTOMER_ID = c.CUSTOMER_ID
JOIN VEHICLE v ON r.VEHICLE_ID = v.VEHICLE_ID
JOIN RENTAL_TRANSACTION rt ON r.RESERVATION_ID = rt.RESERVATION_ID
LEFT JOIN INCIDENT_REPORT ir ON r.RESERVATION_ID = ir.RESERVATION_ID
WHERE r.STATUS = 'Completed'
    OR r.STATUS = 'Active'
ORDER BY rt.ACTUAL_RETURN_DATE DESC, r.RESERVATION_ID;
```

QUERY RESULT

ATION_ID	CUSTOMER_NAME	VEHICLE	RETURN_DATE	INCIDENT_TYPE	DESCRIPTION	DAMAGE_COST	FEE_AMOUNT	REFUND_AMOUNT	NET_CHARGE	ORIGINAL_RENTAL_COST	FINAL_BILL	RESOLUTION_STATUS
1	2 Michael Brown	Mercedes C-Class	26-09-14	Damage	Minor scratch on rear bumper	150	150	0	300	847.5	1147.5	Resolved
2	5 Sarah Miller	Tesla Model Y	26-08-26	Late Return	Returned 3 hours late	0	75	0	75	565	640	Resolved

What We Learned Building It



Normalizing without redundancy

Designing the reservation system to avoid duplicate or inconsistent data across related tables.



Modeling branch-to-branch transfers

Building a logistics table that could represent a transfer between any two branches, not just fixed routes.



Managing shared Oracle privileges

Carefully granting access so every group member could create and modify tables without conflicts.



Balancing complexity with performance

Learning how CHECK constraints and foreign keys stop bad data before it ever enters the system.

Every Requirement, Met

15
Tables Implemented

150+
Sample Data Rows

10
Business Queries

3NF
Fully Normalized

VERIFICATION QUERY

	TABLE_NAME	ROW_COUNT
1	BRANCH	10
2	VEHICLE_CATEGORY	10
3	MANUFACTURER	10
4	VEHICLE	15
5	CUSTOMER	12
6	CUSTOMER_LICENSE	12
7	RESERVATION	12
8	RENTAL_TRANSACTION	2
9	PAYMENT	9
10	VEHICLE_TRANSFER	10
11	VENDOR	10
12	PARTS_PROCUREMENT	10
13	WISHLIST	12
14	INCIDENT_REPORT	10
15	LOYALTY_TIER	4

Live row-count query confirming every table is populated in Oracle.



Thank You

Questions about our design, schema, or queries?